Mind the Badge: How Visualization Badges Impact Reading Visualizations

Valentin Edelsbrunner, Jinrui Wang, Min Chen, Pierre Dragicevic, Benjamin Bach

Abstract—This paper assesses how visualization badges—small informative visual labels placed alongside visualizations—affect people's ability to read, interpret, and learn from visualizations. Similar to visual labels on product packaging, visualization badges highlight potentially relevant information to the reader, such as pointing to a 'Major Finding', cautioning a 'Truncated Axis', or informing about 'AI Involvement'. We scrutinize 10 qualities of badges through three complementary studies: (1) a controlled Prolific study with a general audience (N=24); (2) interviews with 12 domain experts experiencing badges as part of professional visualization applications; and (3) interviews with 5 visualization designers gauging the feasibility of badges for themselves and their audiences. We found clear evidence for badges attracting curiosity, aiding interpretation, fostering trust, and motivating broader adoption. We conclude by discussing the role of badges in data and visualization literacy and exploring their potential to become an accepted convention for visualization. By providing our study materials, reusable badge implementation, and additional badge resources at https://vis-badges.github.io, we invite the community to use, adapt, and further study visualization badges.

Index Terms—Data Visualization, Communication, Transparency, Visualization Badges

1 Introduction

Visualization Badges are small graphical labels displayed alongside data visualizations, designed to help readers interpret and better understand the visualization [21] (Fig. 1). Inspired by product packaging labels [26, 64], visualization badges are designed to help visualization authors disclose design justifications and show honesty about the visualization creation process. For example, readers can be informed about potential deceptions [51] (e.g., *Axis Truncated*), data availability (e.g., *Open Data*), or analysis methodologies (e.g., *Outliers Removed*) (Fig. 6). By communicating this information, visualization badges aim to foster trust in the visualization [42, 50] and the messages it conveys, while ultimately aiming to help readers engage more critically with data and visualization. Compared with purely text-based footnotes [30] or annotations [40], visualization badges also aim for more visual saliency: they can express categories through colors and icons (such as a warning triangle, or a check-mark), provide more concise information through short labels (e.g. *Main Finding*), reveal in-depth information on mouse-over, and allow drilling down into explanations.

Nevertheless, as a concept introduced only recently [21], it is unclear whether displaying visualization badges truly fulfills these aspirations, and what the most effective ways are to design and apply such badges. For example, evidence from studies on product packaging labels suggests that short labels are more effective than long text [64], label placement guides attention [23], and even small amounts of relevant information foster people's inferences about a product [64]. However, it is also not clear if short labels are enough to interpret potentially complex concepts around visualizations and data; if providing too much additional context, may make readers less likely to engage with it; or if showing too much information may reduce trust [42]. Visual badges also add further visual elements to potentially already complex visualization displays, such as dashboards or visualization atlases [63]. Badges hence may increase visual clutter and may reduce focus on the visualization itself [2]. Eventually, there is some evidence [21] that visualization authors can be afraid of adding too many badges, fearing it would reduce trust in the visualization and data analysis too much.



Figure 1: Example visualization badges, colored by their intent [21].

This paper offers a first and critical account of understanding the effect of visualization badges on readers and their potential to improve reading visualizations. To that end, we describe and study 10 *qualities* of badges through three complementary studies. The qualities are informed by how people would experience and use badges (Section 2): first, we investigate whether people do pay attention to badges alongside a visualization, i.e., their visual SALIENCY, whether they spark CURIOSITY in exploring more information or whether they are perceived as visual CLUTTER. Then, we ask how people use badges to understand a visualization, i.e., do they seek further information (ENGAGEMENT), do they understand (COMPREHENSION) and interpret (INTERPRETATION) the labels, and whether labels increase TRUST in the visualization. Finally, we seek a first set of evidence for higher-level goals of badges, whether badges help people learn new CONCEPTS about data, whether readers would become more CRITICAL in interpreting visualizations, and whether people could imagine visualization badges becoming a more widely used convention (CONVENTIONALITY).

Each of our three studies employs a different method and looks at a different subset of these qualities. Study 1 (S1) is a controlled online study (N=24) comparing visual badges with text-based footnotes (Section 3). Study 2 (S2) is an interview study with 12 domain experts who encountered badges as part of professional visualization applications (Section 4). Study 3 (S3) consists of interviews with 5 visualization designers to assess the feasibility of badges for themselves and their audiences (Section 5). Across all studies, we collected both quantitative and qualitative data, which we summarize in a combined scoring system expressing support for each of the 10 qualities.

From the combined results of S1, S2, and S3, we found clear value in using badges alongside visualizations. The visual language of badges appears familiar and easy to understand; badges are valued as bridges to the data, as lightweight aids to interpretation, as a tool to build trust and as a way to make unfamiliar visualizations more accessible. However, we also found that badges can clash with the design of a visualization, may overwhelm readers when overused, could be used in misleading ways, and need to be contextualized and tailored to their bespoke audience to achieve these positive effects. By providing our study materials, reusable badge implementation, and additional badge resources at https://vis-badges.github.io, we encourage the community to use, adapt and further study visualization badges.

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2 Research Questions and Study Rationale

This section describes 10 *suggested qualities* of badges and the associated research questions that motivate our studies. Some of the questions and qualities we investigate are motivated by our personal experience in developing the concept of visualization badges and from co-designing, discussing, and applying specific badges with domain experts across three application cases in climate mitigation, climate communication, and peace analytics [21]. Badge qualities and our research questions are further motivated by the literature on perception and visual attention, context, framing, and communicative visualization, trust and credibility, and visualization literacy and learning. We group our research questions related to badge qualities into three consecutive stages, according to how people perceive and engage with badges: *do they notice badges and are badges considered alongside a visualization (attention)? Are badges used to understand and interpret a visualization (usage)?* and *do badges help learn beyond the use of a single visualization (learning)?* These stages loosely align with existing framings of how people make sense of unfamiliar visualizations [46] (encountering, constructing a frame, exploring visualization, questioning the frame, floundering when reading a visualization).

2.1 Research questions and suggested badge qualities

2.1.1 Stage 1: Paying attention to badges (Attention)

Studies have shown that people form a reliable opinion of the appeal of an infographic very quickly, strongly based on colorfulness and visual complexity [31]. Similarly, Lee et al. [46] found that people's first contact with unfamiliar visualizations was often shaped by emotions bordering on overwhelmedness (*"a really crazy chart"*) while focusing primarily on visual cues, such as *"a lot of rectangles and sort of squares"* as well as *"a bunch of colors"*, and perceived complexity. That said, we still know very little about how people outside traditional data-analysis communities experience and engage with data and visualizations [32] and hence we do not yet know to what extent badges are perceived and understood, or whether they are ignored or even cause irritation. Therefore, we investigate the following potential qualities:

Q1.1 **Saliency**—*Do people notice badges alongside visualizations?*

Q1.2 **Curiosity**—*Do people become curious about badges, as a prerequisite for reading and engaging with them?*

Because badges add visual elements to potentially already complex visualization displays with multiple complementary visualizations and potential interactions (e.g., dashboards, interactive visualization software, data comics [4], visualization atlases [63]), badges need to be designed and applied carefully; making badges too salient will cause visual clutter [42] while making them not salient enough may cause people to miss them [33]. While Edelsbrunner et al.'s [21] design space suggests that badges can be shown at different levels of detail, from full to highly reduced *mini-badges* (Fig. B1), too many mini-badges will increase the number of visual elements on the display. Visual clutter has been mostly studied *within* a visualization, e.g., as a quality metric for information visualization [7] or to create clear and communicative visualizations (e.g., [43, 61]). Studies showed that de-cluttering a visualization leads to higher perceived rating of professionalism [1]. Other studies showed that visual embellishments can improve memorization and recall of a visualization, including its content [5, 10]. However, visualization badges are no *internal embellishments* to the visualization itself, rather, they belong in the category of *external information*, like footnotes [30] or other textual annotations [33, 35, 56], we ask:

Q1.3 **Clutter**—*Are badges perceived as cluttering a visualization display? What is a limit to the number of badges shown?*

2.1.2 Stage 2: Using badges while reading (Usage)

A growing body of literature investigates the context provided alongside a visualization, e.g., through text such as title, labels, captions, descriptions [59, 60]. For example, titles and the framing [44] influence which message a reader first sees in a visualization. Additional contextual information about the topic of a visualization has also been shown to be important for its understanding and interpretation [39]. Annotations can also guide readers towards important insights in a visualization [37]. Badges could have a similar role, e.g., if a badge clearly marks a visualization as a *Main Finding* or when it alerts the reader of *Incomplete Data*. However, work from Burns et al. [14, 15] quantified the effect of contextualizing a visualization with metadata (such as data source or information about the authors) and showed that it did *not* change the perception of relevance, accuracy, clarity and completeness compared to visualizations without extra metadata. Similarly, readers may not find visualization badges relevant. Yet, research on product labels suggests that even small information cues can be highly informative [64]. A similar dynamic may apply to badges: even a single badge, if sufficiently salient and interpretable, may improve how readers interpret a visualization and how much they trust it (e.g., perceived credibility [57], trustworthiness [22], or usability [11]). Compared to static text (e.g., footnotes), badges can support such a *drill-down* mechanism through interactivity [21]. In particular, badges can act as entry points to layered information that uncover more detailed explanations on-demand, and, when appropriate, trigger further interaction. Building on these considerations, we start to examine how badges are used:

Q2.1 **Engagement**—*Do readers read badges and consistently interact with badges to access more detailed information?*

Q2.2 **Comprehension**— *Do readers understand the visual language of badges (shape, icons, and color) as well as their communicative intent?*

Badges may also help readers avoid common pitfalls, often discussed as deceptive visualization techniques [45, 51], for instance by making it easier to notice the use of a *Non-Linear Scale* or the presence of an *Inverted Axis* [24, 27]. In doing so, badges may support not only *visualization literacy* [12], but also the *liberation* of authors' design choices: rather than avoiding potentially misleading, yet sometimes beneficial, encodings by default, authors could use them deliberately and affix a badge to help readers recognize them. In this sense, badges have the potential to serve as lightweight, in-situ hints that not only warn about pitfalls, but also guide, inform, and contextualize. We ask:

Q2.3 **Interpretation**—*Is badge information taken into account when reading and interpreting a visualization, i.e., does the presence of badges prevent misreading a potentially deceptive visualization and does it support more accurate data interpretation?*

Trust is an important factor in how visualizations are received and acted upon by readers [50]. It is, however, a multi-dimensional construct and difficult to measure, as there is no clear consensus on a definition of trust in visualization contexts [22]. Trust is often discussed in relation to constructs such as credibility, clarity, reliability, familiarity, and confidence [22, 49, 50]; badges may also be entangled with these constructs. In a sense, badges are not neutral add-ons: when attributed by the author, they can act as, e.g., *credibility signals* that readers may interpret as evidence of author accountability [37]. Yet such signals are only as trustworthy as the process behind them. This creates a risk of *miscalibrated* trust if badges are self-asserted or poorly substantiated, but also an opportunity to *calibrate* trust when badges are grounded in transparent practices such as disclosing limitations in the data; or by disclosing assumptions. We therefore ask:

Q2.4 **Trust**—*Do badges increase a reader's trust in data, analysis, and visualization, e.g., by disclosing confirming information or by acting as 'attention roadblocks'?*

2.1.3 Stage 3: Learning for visualization literacy (Learning)

This last group of aspects describes possible cognitive effects and the educational potential of badges and whether exposure to them can contribute to improved data and visualization literacy. Our questions are deliberately pushing the boundaries of what we can assess, therefore aimed at providing an initial understanding to inform future studies.

The ability to read, reason about, and reflect on visualizations [12] has been studied through visualization literacy tests such as VLAT [47, 52] that operationalize literacy as chart reading; or by testing critical thinking of people's ability to read (misleading) charts [27]. Yet relatively little work asks how we can foster such literacy while engaging with a visualization [3, 55]. Badges may act as such a lightweight

in-situ learning opportunity, by surfacing important information and offering on-demand explanations. Badges might support opportunistic learning [48] by pointing to additional information (e.g., explaining how to read a logarithmic scale or a new visual encoding technique):

- **Q3.1** **CRITICALITY**—*Do badges promote critical reflection on data, analysis, and visualization?*
- **Q3.2** **CONCEPTS**—*Do badges help readers learn about previously unknown concepts in data analysis and visualization?*
- **Q3.3** **CONVENTIONALITY**— *Does experiencing badges motivate their adoption and foster expectations for better transparency conventions in visualizations?*

2.2 Study methodology

Studying visualization badges is not straightforward for a number of reasons. First, whether and how badges can help reading visualizations may strongly depend on readers' needs, literacy, prior expectations, and motivations. This is reflected in related work on product packaging labels [16, 64], which shows that factors such as domain knowledge, awareness, familiarity, skepticism, and motivation shape how consumers draw inferences from labels. Second, the communicative intent of badges can shift from one context to another—for example, *Outliers Removed* may function as a confirmation or a warning [21]. Finally, the absence of appropriate control conditions and the complexity of establishing standardized measures, such as the multifaceted nature of trust in visualizations [22], add to the challenges in evaluating badges. Consequently, to address our research questions, we designed three complementary studies S1-S3, set in different environments and focusing on different but overlapping subsets of research questions.

S1 **General audience comparing badges to footnotes**—Our first study compares visualization badges to a well known baseline technique: textual footnotes. The study targets a general audience (run on Prolific), with varying familiarity with visualizations or data analysis. It is also meant to provide some first evidence for *some* of our research questions and help inform the two following studies. The comparison provides first answers on the following badge qualities: SALIENCY, CLUTTER, ENGAGEMENT, COMPREHENSION, INTERPRETATION, TRUST, and CONVENTIONALITY of badges.

S2 **Badges deployed to domain experts**—This study examines badges in two web-based real-world visualization projects with the goal of understanding all 10 potential qualities in more contextualized settings and with domain experts acquainted with data and visualizations: a public visualization atlas [53] and visualizations targeted at peace builders and researchers in peace studies [62]. Across both projects, we interviewed 12 professional participants (researchers, analysts, policy makers, journalists).

S3 **Visualization designers' viewpoints**—To gauge the applicability and conditions of using badges with experts in visualization, we interviewed 5 professional visualization designers about how they thought badges would affect readers in terms of COMPREHENSION, INTERPRETATION, TRUST, CONCEPTS and CONVENTIONALITY as well as whether they would use badges in their own projects.

2.3 Data analysis

We devise a coding scheme to compare and capture trends across studies. For S1, these codes are directly based on answers from Likert scale questions (Figs. 4 and 5). For S2 and S3, the codes have been assigned by two independent coders and resolved through discussion. For intercoder agreement analysis for S2 and S3, see Secs. B.6 and C.2. To devise the support scheme, we first coded the answers given by each participant for each badge quality into one of the four categories.

- **Negative evidence**: the participant expressed negative evidence with respect to the badge quality, for example that badges are *not* perceived as salient or that they do *not* promote criticality. In S1, this code was applied to all participants who answered *disagree* or *strongly disagree* on the Likert scale.
- **Positive evidence**: the participant expressed positive support with respect to the badge quality, for example that badges are perceived as salient or that they foster trust. In S1, this code was applied to all participants who answered *agree* or *strongly agree* to the respective Likert-scale item.
- **Mixed evidence**: in S2 and S3, this code was used when a participant made *both* positive and negative comments about a badge quality, for example by indicating that the badge quality applies only sometimes or that the quality depended on a specific context or case, e.g., *"badges are salient if . . . "*. In S1, this code was used for participants who selected *neutral* on the Likert scale.
- **No data**: this code was used only in S2 and S3 when no clear answer could be obtained, for example because participants did not respond meaningfully to the question (e.g., *"nothing comes to my mind"*) or because the respective quality did not come up during the interview.

To better communicate the results for these codes during our reporting in the remainder of the paper, we created a small signature visualization with three squares and three colors:  . The left **red square**'s saturation encodes the relative share of participants voicing negative evidence, the middle **yellow square** shows mixed evidence, and the right **blue square** visualizes positive evidence; the numbers inside the squares show the corresponding counts of participants expressing that quality. Answers coded as *No data* were excluded from the visualization. This encoding results in a specific signature visualization for each badge quality and study. For example, a badge quality such as SALIENCY with all participants expressing positive evidence would result in a strong blue square  ; a badge quality where half of the participants provided negative evidence and half provided positive evidence would be shown with both light blue and light red squares  . A badge quality with participants equally split across the three categories would appear as three light squares  .

For each study S1, S2, and S3 we report the methodology and findings individually and how the results add to each other (Sections 3, 4, 5), using the same coding scheme. Section 6 then uses this shared scheme to synthesize findings across all three studies. For all three studies reported in this paper (S1 – S3), ethics approval was obtained from Inria's COERLE ethics committee; study materials, results, and implementations are provided in the supplementary materials available at https://vis-badges.github.io.

3 STUDY I: COMPARING BADGES TO FOOTNOTES

This study compares visual badges to textual footnotes and aims to provide first evidence on SALIENCY, CLUTTER, ENGAGEMENT, COMPREHENSION, INTERPRETATION, TRUST, and CONVENTIONALITY.

Stimuli. We selected two visualizations with original footnotes from public, trustworthy sources (Figure 2). Both visualizations addressed climate change, grounding the study in a familiar topic. We chose one line chart[1] (Vis-1) and one bar chart[2] (Vis-2) because they are among the most common and well-known chart types [6], are commonly used in public communication, and were reasonably designed to not puzzle participants with visualization-reading questions (see below).

Study Design. We conducted an online study in two parts (**S1a** followed by **S1b**). Across conditions, the visualization part of the stimuli remained the same (Fig. 2). However, in S1a, participants completed a between-subjects study in which each group saw either the FOOTNOTE or BADGE condition, first on Vis-1, then on Vis-2. Approximately ten days later, the same participants were invited back to complete the second part S1b where they viewed the condition they had not seen during S1a. They were then asked to directly compare both conditions, seen side by side with alternating left-right positions of the stimuli. The two study parts—S1a and S1b—were the result of an evolving study design; S1a aimed at gathering *qualitative* data to understand how each technique was perceived individually, i.e., without exposing participants to the respective other condition. However, this method proved inconclusive due to the novelty of badges. Hence, S1b adopted a direct and *quantitative* comparison.

Conditions. In the FOOTNOTE condition we used the original footnotes from the original source, for the first part of this study S1a. In the BADGE condition, we replaced these footnotes with equivalent

[1] https://ourworldindata.org/co2-emissions
[2] https://www.carbonbrief.org/

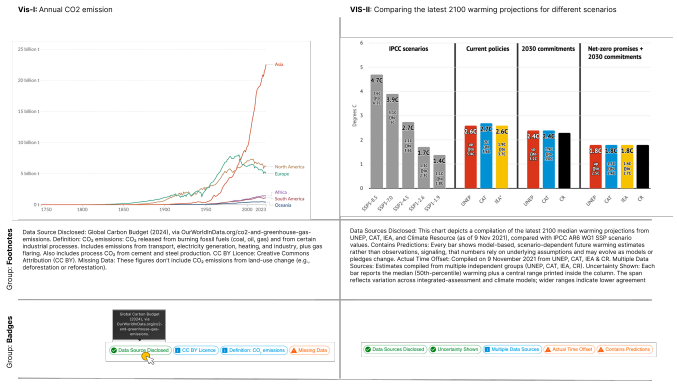


Figure 2: Stimuli Vis-1 and Vis-2 used during S1, shown with either footnote (middle row) or badge condition (bottom row). Hovering a badge reveals a description.

interactive visualization badges: for each footnote, we created a badge featuring a label and a description. The description was revealed on mouse over as a small pop-up over the label. While the description matched the footnote text, the label was a short version of it to fit into the badge. For example, where the footnote provided a link to the data source, the badge label was *Data Source Disclosed*, while the description pointed to the URL of the data source. As illustrated in Fig. 2, this resulted in four badges for Vis-1 and five badges for Vis-2, each derived from the author-provided footnotes. For the second part of this study S1b, where footnotes and badges were compared by participants, we also added the badge labels to the footnotes and updated the footnotes to match the exact same information as the badge version to allow a fair comparison (Fig. 2).

Piloting. We conducted a pilot study with 5 participants. Based on participant feedback, we made minor adjustments to the wording and interface design, while keeping the overall procedure unchanged.

Participants. In S1a 24 participants took part (3 via university mailing lists; 21 via Prolific), compensated at 15.61£/hour (median 19 minutes). All S1a participants were invited to S1b, and 18 returned (9 footnote-first; 9 badge-first), compensated at 9£/hour (median 15 minutes). We recruited English speakers with at least an undergraduate degree. For full demographics, see Sec. A.1 and Sec. A.2.

Procedure. S1a and S1b were implemented using the ReVISit framework [18,20] and conducted as web-based studies requiring a minimum viewport of 1300 × 750 px. Participants in **S1a** were assigned to either a badge condition or a footnote condition. They then completed the same task for the two stimuli, presented in randomized order (Fig. 2). The participants task was: *"Imagine you're presenting this visualization to your boss. Write down the text you would use for your speech"*. This task was designed to engage participants with the visualization and its meaning and have them express their interpretations, while avoiding nudging them towards the footnotes or badges. Participants were neither onboarded to badges nor given explicit instructions to interact with them. After completing both tasks (i.e., writing two speeches), participants answered a set of open-ended questions about whether they noticed footnotes/badges and whether they used them during the task, and provided feedback on helpfulness. They also completed a 5-point Likert questionnaire (Strongly Disagree–Strongly Agree) covering SALIENCY, CLUTTER, INTERPRETATION, COMPREHENSION, TRUST and CONVENTIONALITY. Finally, participants completed a demographics questionnaire (Sec. A.2).

For **S1b**, after approximately ten days, we re-recruited S1a Prolific participants for a follow-up direct comparison. Of the initial 24 participants in S1a, 18 returned for the second part, S1b, with 9 from each S1a group. Participants first explored the stimuli from the other condition for at least 60 seconds per visualization (with *Next* disabled until 60 seconds had elapsed). They then compared the badge and footnote versions side by side and answered the following two questions per stimulus: *(1) For understanding this visualization yourself, which version do you prefer?* and *(2) For presenting and explaining this visualization to others, which version would you prefer to use?* Participants selected their preference shown in Fig. 3, as well as explained their decisions with written explanations. Finally, participants completed the same Likert questionnaire as in S1a, but rated both footnotes and badges for a direct comparison (Fig. 5).

Analysis. Badge interaction logs were recorded while participants conducted the study only for S1a; we recorded the number of hovers and time spent on badges, and report them descriptively. Likewise, Likert-scale responses are reported descriptively. Participants' written speeches were read by three authors of this paper and then discussed; quotes were selected as representative examples, capturing nuances.

3.1 Findings

We organize our analysis around (i) comparing badges and footnotes and (ii) our research questions and badge qualities from Section 2.

3.1.1 Comparing badges to footnotes

In S1a, **badges and footnotes were comparable overall** (Fig. 4), implying that badges could be a viable alternative to footnotes. Badges were rated somewhat higher for INTERPRETABILITY and TRUST, whereas general usefulness and CONVENTIONALITY were rated equally. In the open responses participants wrote, FOOTNOTES were cited slightly more often (11/12 participants) than BADGES (8/12). Participants cited both footnotes and badges as useful, using footnotes mainly to clarify terms and context. Badges however, seem to have evoked some deeper engagement with the data, through *"explain[ing] how the data could be read"*[P14], e.g., *"badges [were] important to determine the accuracy and reliability of the data."*[P1] Badge condition participants also reported critically on themselves, *"[badges] made me aware of the limitations of the chart."*[P2] and that *"I emphasized the uncertainty [in my description]."*[P2]

In S1b, where participants directly compared badge and footnote versions, **support for BADGES was stronger**. BADGES seem to be the overall preferred choice by participants (Fig. 3) with a clearer preference for scenarios that require understanding a visualization. Asked which condition they preferred for presenting a visualization, the decision was less clear, with people saying that *"[I prefer] footnotes [for] presenting to others because they provide complete explanations that I can refer to."*[P8] Nevertheless, BADGES were also ranked higher on all but one criterion (Fig. 5): they were perceived to cause less CLUTTER, had easier INTERPRETABILITY, easier COMPREHENSION, and were preferred for CONVENTIONALITY (Fig. 5). Participants often explained their preference for BADGES in terms of organization and readability: *"Badges group the information into categorized sections, making it easier to identify important information, read and interpret it than the large block of text in the footnote version."*[P13] In summary, badges appear to be a promising alternative to footnotes, providing a clearer overview of available information and supporting interpretation. At the same time, these findings are based on a small sample and should therefore not be read as conclusive evidence.

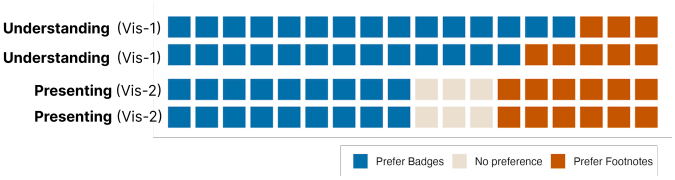


Figure 3: Participants' preferences S1b for badges vs. footnotes when understanding the visualization (top rows) and presenting it to others (bottom rows), per stimuli. Each square represents one participant.

3.1.2 What this study tells us about badges

For this study, we report on two overall coded schemes (Sec. 2.3): **S1a at the top**, derived from Fig. 4, and **S1b at the bottom**, derived from Fig. 5, because both sub-studies contribute to the findings below.

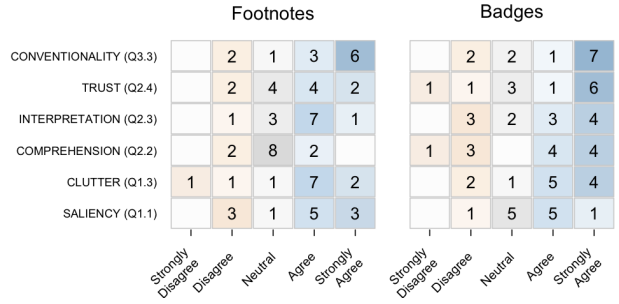


Figure 4: Likert ratings in **S1a** for footnotes (n=12) and badges (n=12).

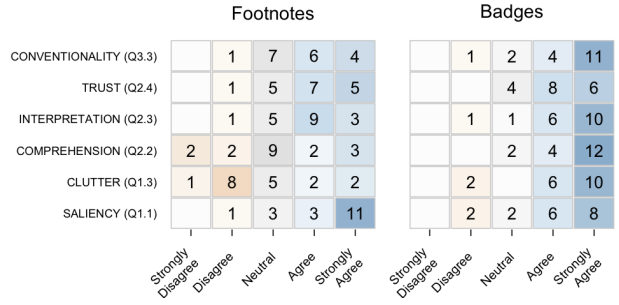


Figure 5: Likert ratings in **S1b** comparing footnotes to badges (n=18).

1. Attention.

1 5 6 2 2 14 Saliency (Q1.1). In S1a, 5/12 participants had a neutral stance, while 6/12 agreed or strongly agreed that badges were easy to spot; in S1b, this increased to 14/18. *"Yes, I noticed them immediately."*[P20], *"I noticed this towards the end, and found it very helpful."*[P14], but also *"I had not noticed that they were interactive- they'd not given the indication that they were."*[P17]

2 1 9 2 16 Clutter (Q1.3). In S1a, 9/12; in S1b, 16/18 participants did not perceive badges as cluttering the visualization display. Participants described badges as *"well-designed, not getting in the way, but there when I was interested."*[P2] as well as *"not [...] cluttered [but looking] cleaner and more organised than [footnotes]."*[P15]

2. Usage.

11 13 Engagement (Q2.1). In S1a, badge-hover activity was logged during the main tasks of this study; badges elicited engagement but it was bimodal across stimuli: only 4/12 participants hovered over any badge on Vis-1, whereas 9/12 did so on the more complex Vis-2, which included jargon and uncertainty ranges. *"The presence of interactive or detailed elements, like hover descriptions on badges, significantly deepens engagement by transforming a static chart into a self-documenting tool."*[P20] *"I might have used the badges if I was more aware of them."*[P5]

4 8 2 16 Comprehension (Q2.2). S1a shows a light split, 4/12 did not understand badges. In S1b however, most participants (16/18) reported understanding the visual language of badges. *"The titles alone, such as Contains Predictions, were completely clear for quick context. The hover explanations for Uncertainty Shown and Multiple Data Sources offered crucial, non-optional detail for a thorough knowledge of how the uncertainty was assessed and modeled."*[P20] *"Easy to understand, but maybe not as helpful as it might have been without hovering over."*[P17]

3 2 7 1 1 16 Interpretation (Q2.3). In S1a, 7/12; in S1b, 16/18 participants agreed that badges are helpful for understanding the visualizations. However in S1a (3/12) reported that badges did not help their understanding of the visualization. *"I noticed [badges] but didn't actively use them or felt they were relevant to what I was looking for."*[P21] *"They did not help me especially, as all I got from them was [meta information]."*[P17]

2 3 7 4 14 Trust (Q2.4). In S1a a slight majority (7/12) reported, that badges supported trust in the visualization; this increased to 14/18 in S1b. One participant noted: *"I liked having them, it made me trust the information more!"*[P11]

3. Learning.

2 2 8 1 2 15 Conventionality (Q3.3). Most S1a (8/12) and even more S1b (15/18) participants agreed, that badges should become a common practice. *"The badges have won me over. I don't see any reason to go back to using endnotes or footnotes in scientific communication unless the media are static such as in paper and cardboard. For computers and the www badges do everything better. They have the same information quantity and quality, they offer better organization, and they allow researchers to make better use of space which is a scarce resource for infographics."*[P6] However, participants also emphasized that badges need to be tailored to their audience, to *"make it easy for a normal person to understand the information represented."*[P16]

3.2 Summary of S1 and Rationale for S2

S1 shows that readers preferred Badges over Footnotes in a controlled comparison and provides first evidence that badges are Salient, add little Clutter, support Interpretation, foster Trust, and are considered appropriate for broader integration into other projects (Conventionality). At the same time, the setup of S1 does not reflect how people typically encounter visualizations in real-world settings. Users rarely encounter visualizations as isolated charts; instead, they scroll through web applications or navigate dashboards, often with multiple views, considerable text, and interaction affordances that shape how information is experienced. This motivates S2, where badges are evaluated in the setting for which they are ultimately intended.

4 Study II: Badges Deployed to Domain Experts

This study scrutinizes badges in the context of real-world visualization projects used by professionals and is designed to address all 10 research questions (Sec. 2.2). For this study, we collaborated with two web-based visualization projects that target different audiences and employ different forms of visual presentation (Fig. 6). *The UK Co-benefits Atlas* (Atlas) is a multi-page web-based visualization atlas with dense, dashboard-like pages combining around 20 visualizations and explanations per page, and hundreds of pages with data for climate decision-making (Sec. B.1.1). *PeaceRep* (PeaceRep) comprises several web-based visualization projects on peace processes, often visualizing uncertain data in a sensitive domain that requires bespoke approaches to communication, transparency, and information disclosure (Sec. B.1.2).

4.1 Study Design

We conducted this study in three stages: (1) badge selection, (2) badge assignment, and (3) stakeholder evaluation. This structure reflects how badges would realistically be introduced into a visualization project.

Stage 1: Badge Selection. As a prerequisite for this study, we required a set of badges to add to the projects. To this end, we held two two-hour badge-selection workshops, one with the Atlas team and one with the PeaceRep team; importantly, participants were involved in creating the respective project(s) (see demographics in Sec. B.2). In the first hour, participants individually assigned badges to their projects; in the second hour, these selections were discussed collectively. For more details we point to Sec. B.2.

Stage 2: Badge Assignment. The second prerequisite was to affix badges to the live visualization implementations. To do so, we developed a JavaScript library that operationalizes the badge design space introduced in our prior work [21]. The library supports different badge placements, visual modes, and interaction affordances, which we used to implement a *badge priority* system, which was inspired by the two workshops: salient round badges for high-priority information, standard badges for regular use, and mini badges that expand on hover to reveal the badge label (Fig. B1). For more details see Section B.3.

Stage 3: Stakeholder Evaluation. Eventually, we conducted 12 semi-structured one-to-one interviews with domain experts, each lasting around one hour. For recruitment, we asked our collaborators to connect us with people who either used the projects in their professional work (n=4) or worked in closely related fields and could therefore be considered potential users (n=8). Based on the demographics reported

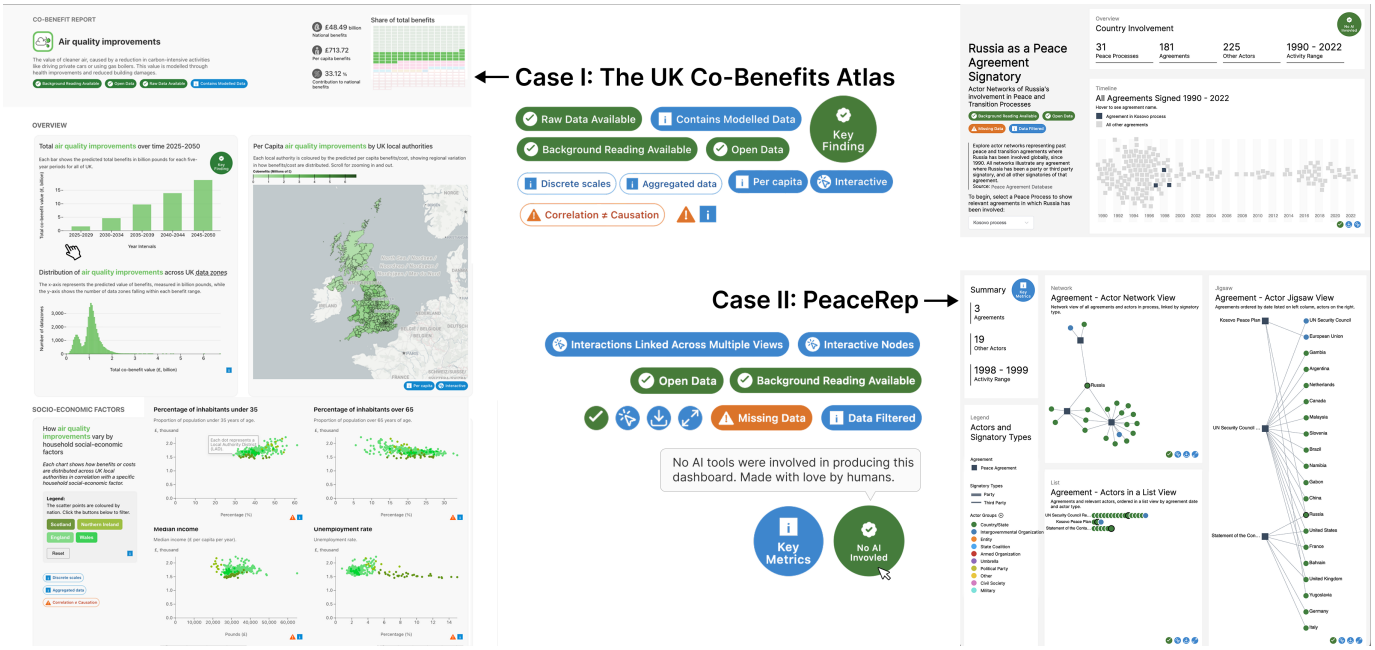


Figure 6: Two example visualizations used during S2, with badges applied to one UK Co-benefits Atlas page (left) and one selected PeaceRep example (right). For a full list of all projects used during S2 we refer to Appendix Sec. B.5

in Section B.4, participants came mainly from academia (n=7), with additional participants from community organizations (n=2), local government (n=1), the private sector (n=1), and healthcare (n=1). Their roles included managers (n=4), researchers (n=3), analysts (n=2), and one participant each working as a journalist, consultant, and communicator. Depending on the background, participants engaged either with the ATLAS (n=9) or the PEACEREP (n=3) (Section B.5).

The lead author of this paper conducted 12 online interviews, all of which were recorded and transcribed. Interviews began inquiring about the participant's professional role and their familiarity with the visualization project. Participants were then invited to freely explore the projects (ATLAS or PEACEREP) while sharing their screen and thinking aloud for around 15min. Importantly, and like in S1, **visualization badges had not been previously introduced to participants and were not mentioned in the beginning of the study either**. Our 10 research questions from Sec. 2.2 directly informed the questions in the concluding interview, which lasted for another 45 minutes.

4.2 Analysis

We analyzed the interviews using directed content analysis [36] to code each interview against our 10 research questions (Sec. 2.1). We employed a participant-by-dimension matrix in which each cell received one predefined code including one optional adjacent cell in the matrix, to justify codes, as well as to capture valuable decisions and nuances that were not reflected in the coding matrix, allowing for a light inductive analysis [25]. For example SALIENCY was coded with `YES/UNCLEAR/NO`. See all codes in Tab. B3.

Two authors of this paper coded all transcripts independently. For details on intercoder agreement analysis, see Sec. B.6. The coders then reviewed disagreements dimension by dimension, returned to the transcripts, and discussed ambiguous cases until reaching a shared interpretation, which we used to report on the findings below.

4.3 Findings

We now report findings from the 12 interviews. Using our final coded matrix, we employ the same scheme introduced in Sec. 2.3.

1. ATTENTION.

1 10 SALIENCY (Q1.1). A clear majority of participants in S2 (10/11) noticed badges, compared to only 6/12 and 14/18 in S1a and S1b respectively. Moreover, badges in S2 were placed in much more complex visualization displays (Sec. B.5) but were adapted to different styles and participants spent more time engaging with the task (Fig. B1). Badges placed at the beginning and high on pages, for example in the ATLAS project, were often the first thing users commented on when opening the ATLAS, because people were *"instantly [. . .] drawn to them"*[P7], or because the round *Key Finding* badge immediately signaled *"right, here's my focus"*[P1]. These **badges were often read as page-level signposts and as markers of important takeaways**. Yet, noticing the badge system did not mean that all badges were equally salient: some participants noticed only some badges, while others noticed badges without reading them because they did not seem relevant to them.

9 CURIOSITY (Q1.2). CURIOSITY was not evaluated in S1 and was only assessed here; 9/12 participants expressed that badges sparked, or could spark, curiosity, while the remaining three were excluded from this count because they were coded as *No data* (Sec. 2.3). Badges sparked curiosity through (i) their visual language: *"I want to go and click on them all."*[P1] One participant said that the *" warning signs [...] make me curious about, okay, what am I being warned of here?"*[P9] and (ii) through their information, **if** the reader cared about it, e.g., P10 commented on the "Background Reading Available" badge that *"I immediately want to go to see what the background reading is."*[P10].

2 6 4 CLUTTER (Q1.3). Similar to the findings in S1, participants did not experience badges as a major clutter concern in the visualization projects we presented them with, describing badges as*"neatly put there."*[P5]. While this may reflect the care we put into designing these badges for these real world projects and their users, asked about badges potential to clutter more generally, half of the participants (6/12) raised potential concerns. This was especially the case when badges were perceived as redundant or not helpful, e.g., when the same badge was repeated across multiple charts in the ATLAS. Such repetition should be handled by placing the respective badge once *"at the beginning of a section and then as few as possible"*[P8]. When asked about a "good" number of badges per chart, participants agreed that less is more, with many suggesting **around three badges as a reasonable upper limit**, while slightly more were considered acceptable at the page level.

2. USAGE.

3 8 ENGAGEMENT (Q2.1). Without us explicitly introducing badges, a slight majority of participants intuitively engaged with badges (8/11, similar to Vis-1 (4/12) and Vis-2 (9/12) in S1a), describing this interaction as natural, e.g., *"I automatically [...] hover over them to see what they mean."*[P9]. (One participant was coded as *No data*, because of technical issues, this stage could not be observed.) However, badge interaction was also criticized because all badges offered tool-tips on hover, while only some were clickable. For example, "Background

Reading Available" routed to the methodology section, whereas "Key Finding" only revealed a tooltip and offered no further action.

4 8 COMPREHENSION (Q2.2). The visual language of badges was clear and intuitive for most participants; 8 participants reported that badges were easy to understand, while the remaining 4/12 expressed a mixed stance. This reflects well our findings from S1. Badges felt familiar, like in *"[product packaging,] there is a check on any product that is environmentally friendly or green product"*[P12]. Readers inferred that *"there's a color coding going on"*[P9]. However, labels and tooltips were less consistently understood, e.g., one participant said its *"quite academic language"*[P5].

1 3 8 INTERPRETATION (Q2.3). 8/12 participants reported that badges were useful for understanding and could support more accurate data interpretation. Of the remaining four participants, one did not think badges aided understanding, while three expressed a mixed stance, suggesting that badges particularly support interpretation when charts are unfamiliar or visually complex. In general this trend was also visible in S1. Badges can help *"make really complex data accessible."*[P1] One participant described badges as a **bridge to the data**, saying *"this is opening up that next layer down, which is actually access to the data [...] and people are going to be able to understand it"*[P7]. Badges were criticized when they were too generic or redundant i.e, that **if a visualization needs *"lots and lots of extra keys, pointers, arrows, guidance,"*[P6] the visualization may no longer do its job**.

4 8 TRUST (Q2.4). 8 participants reported that badges increased trust in the visualization, data, and analysis process, the remaining 4/12 expressed a more conditional stance. Increase in trust was often associated with transparency (*"understanding where the data comes from"*[P7]), and clear communication about the methodology (*"because this is not trying to hide anything"*[P1]), making readers more confident about what they were seeing. *"I trust it more because [badges are] trying to be as clear and open as possible."*[P1] However, results for S2 show more nuances and criticality than in S1. For example, one participant noted the risk of being too transparent *"if there was [a warning badge] on every single [chart], I'd start to kind of go, 'can I trust this data'?"*[P5].

3. LEARNING.

2 1 8 CRITICALITY (Q3.1). Whether badges can promote critical reflection on data analysis and visualization remains unclear from our data for S2 and was not tested for in S1. This is also reflected in our agreement analysis (Sec. B.6); 4 participants were coded as *no data*. Critical reflection was most often associated with warning badges (such as *Correlation is not Causation*); one participant had explicitly stated, that *"I've just got more critical"*[P5] after seeing such a badge. Interestingly, P3 pointed to a **tension between trust and criticality**, suggesting that if a visualization provides *"more badges, more context and more information,"*[P3] readers may become less critical because this additional information increases trust and may reduce the need to scrutinize the visualization more carefully. This could potentially be exploited by dishonest parties.

1 7 CONCEPTS (Q3.2). 7/12 participants reported that badges could help readers learn previously unknown concepts or unfamiliar ways of reading a visualization. Of the remaining five, 4 were coded as *no data* and therefore excluded; 1 participant did not think badges would be useful for that purpose. Some participants described badges as light-weight in-situ hints that provide entry points for drilling down into more detailed information and help readers develop *"a toolkit for identifying issues"*[P10]. As one participant noted, badges *"could be helpful for learning (1) how to navigate or read a site and (2) what certain practices imply."*[P12]

4 7 CONVENTIONALITY (Q3.3). Similar to S1, there was general support for wider badge use (7/12) with another 4/12 supported conventionalization under certain conditions. One participant was coded as *no data*. One participant reflected on using badges to *"adopt [...] standards to which we will hold ourselves"*[P10] and thereby bind themselves *"to a set of rules, like Odysseus being tied to the mast."*[P10]. More conditional responses, however, did not support badges as a default layer on every project or chart, but tied their use to project complexity and relevance: *"not everything needs [to be] explained,"*[P4] and badges *"really have to give some sort of important message"*[P6]. We return to this issue in the discussion (Section 6).

4.4 Summary of S2 and Rationale for S3

10/12 participants expressed a positive stance toward the concept of badges and their integration into the websites used during the interviews. However, badges still need to be affixed to visualization displays, and designers need to feel that they will benefit their audience before adopting them in their projects; we examine this in S3.

5 STUDY III: VISUALIZATION DESIGNERS' VIEWPOINTS

To further gauge the feasibility of badges, we engaged with 5 visualization designers to reflect on their willingness to use badges in their own projects and their perceived value for readers in their target audience. We provided designers with resources on badges, including our previous paper and badge catalog from [21] as well as the same projects used in S2 (Sec. B.5). Designers could either participate in an asynchronous questionnaire conducted via email (n=3) or in an online interview (n=2), which was guided by the same questions from the questionnaire. The interviews lasted one hour, were transcribed, and the data was coded by the same two coders as in S2. For an overview of participants' professional experience, we refer to Sec. C.1.

5.1 Analysis

As in S2, two authors coded the materials independently using a codebook (Tab. C2), that followed a similar blueprint to that of S2, to obtain one final participant-by-dimension matrix. We analyzed S3 using the same overall approach as in S2 (Sec. 4.2) and coded COMPREHENSION, INTERPRETATION, TRUST, CONCEPTS and CONVENTIONALITY, asking designers how they thought their audience would perceive badges with respect to these qualities, as well as whether they would use badges in their own projects. For intercoder agreement analysis, we point to Sec. C.2. Designers' personal opinions on badges are also reflected in the findings below, but were not coded.

5.2 Findings

We now report findings from 5 engagements with professional visualization designers, using the same coding scheme introduced in Sec. 2.3.

1. ATTENTION. S3 was not designed to evaluate this stage; nevertheless, designers reflected on it (a lot), often with a more **negative stance linked to the aesthetics of a visualization**. As one designer noted, readers *"begin by looking at the elegance"*[P2], and if badges are styled well, they can become *"beautiful artifacts"*[P2] that integrate into a designer's visual language. Concomitantly, badges may *"distract attention from the main visualization"*[P4] and *"may not [be] compatible with different UI styles"*[P4]. P1 expected badges to work well in mainstream news graphics, where their *"vivid coloring"*[P1] could *"draw attention to [badges]"*[P1], but less so *"for more visually focused, design-oriented data works"*[P1]. More generally, designers' main concern was badges *"clashing or interfering with the aesthetics of a design"*[P3], which is why most expressed a need for *"a degree of styling flexibility"*[P2]. In regards to the projects used in S2 one designer who scrutinized them criticized, that *"some labels feel over-explained in practice, while others rely too much on prior knowledge"*[P4].

2. USAGE.

3 2 **COMPREHENSION** (Q2.2). Visualization designers reported their target audience would generally understand the visual language of badges because *"the basic geometries of circle, triangle, square are very universal, and used widely in everyday life."*[P1]. As in S2, however, the **terminology used in the badges depends on audience literacy**. For example, *"'Dual Axis Encoding' that's data visualization lingo, and for many people they have no idea what you mean by this"*[P3], with P2, P4 and P5 having similar concerns, but P2 arguing that badges provide *"the space to even explain those things once people got to that step."*[P2].

4 1 **INTERPRETATION** (Q2.3). Designers were conditional about whether badges aid interpretation. They saw clear value in using badges for less experienced readers or unfamiliar charts, or article-style visualizations e.g., *"applying badges directly to data points that require more*

attention or deeper understanding can be helpful for those who are not that experienced with reading charts"[P3]. Simultaneously, designers were more hesitant to use badges that tell readers what to take away from a visualization. As P2 explained, *"sometimes [the visualization is] there to show you the data and leave it up to you to interpret because everyone's got different entry points and requirements for it."*[P2]

2 3 **TRUST** (Q2.4). Designers generally agreed that badges can foster trust by providing a layer of verification and by making transparent *"what you're showing with the visualization and the choices that have been made, that are normally not visible."*[P1]. However, they also noted that if readers are already sceptical of a project or website, they *"would not be convinced of the opposite because of badges"*[P1].

3. LEARNING.

5 CONCEPTS (Q3.2). Similarly to the findings from S2, designers saw value in using badges as a way to educate their readers about potentially unknown concepts, practices, or ways of reading a visualization. 2/5 designers reported that badges can provide a 'degree of liberation', by making *"experimenting with unusual visualization types easier"*[P1], because badges can be used as a *"coaching assistant available in a nice neat, consistent, established way, then that does give a certain degree of extra confidence [for designers]"*[P2]. In the context of learning, P2 noted, that once participant learned information provided through badges they should disappear, but *"still make it available on demand if it's required as a refresher or if you're a new visitor, but if you're going back every single day [to the the same COVID dashboard], you don't need to tell me [how death is counted]"*[P2].

4 1 CONVENTIONALITY (Q3.3). Compared to S1 and S2, 4 designers were more hesitant about adopting badges, because of what designers reflected on in ATTENTION. Assuming that badges are used more often across projects, P2 and P5 contemplated widely about conventionalization: *"if you don't have the badges, what are you hiding? Why aren't you part of the badge club?"*[P2] *"Some ways to adapt it to the corporate identity are useful, because that way it can feel a little bit more like part of the page. But [...] if it's too subtle, maybe people wouldn't notice it."*[P5] *"It can really enhance visual literacy, because maybe you haven't thought about the fact that you should pay attention if axes are truncated before you've seen this badge. So it can even have an educational effect."*[P5]

6 DISCUSSION AND FUTURE WORK

This research set out to explore if and how visualization badges would affect people's ability to read, interpret and learn from visualizations. To that end, we tested 10 qualities of badges across three complementary studies S1-S3. Below we summarize the main findings and discuss nuances and themes that emerged throughout this qualitative work.

Stage	Qualities	S1a	S1b	S2	S3
		n=12	*n=18*	*n=12*	*n=5*
Attention	SALIENCY				
	CURIOSITY				
	CLUTTER				
Usage	ENGAGEMENT				
	COMPREHENSION				
	INTERPRETATION				
	TRUST				
Learning	CRITICALITY				
	CONCEPTS				
	CONVENTIONALITY				

Table 1: Table summarizing quantified stances across studies.

6.1 Summary findings

Table 1 summarizes stances and opinions across studies, as captured by our coding scheme (Section 2.3). From the amount of saturated blue in these quantified results, we can deduce a **clear perceived value in using badges alongside visualizations**, across all three stages: attention, usage, and learning. **The general audience in S1b were most positive** about badges after having compared them to footnotes. Without such a direct comparison, the difference to footnotes was not strong. In S2 and S3, **domain experts and visualization designers were more nuanced** (yellow=mixed evidence) providing critical thoughts about badge applications which we discuss further in Sec. 6.2.

Generally, we can assess that **badges were noticed by participants** and **were considered** SALIENT. In contrast, badges in our study were not perceived as CLUTTERing in simple environments with a single visualization (S1). In more complex environments (S2), **the repeated use of badges across a website could lead to visual clutter** caused by the repetitive use of badges. Likewise, too many badges could lead to *badge fatigue*—the phenomenon of having to deal with a lot of information and which has already been described in previous expert workshops [21]. General audience (S1), domain experts (S2), and visualization designers (S3) further mostly agreed that the visual language of badges was familiar and easy to understand (COMPREHENSION). Yet understanding a badge's intent also depended on the clarity of its label and description as well as an audience's literacy and context, such as whether readers were first-time or returning users. When readers understood the communicative intent of badges, they valued them as *"bridges to the data"* making underlying data, provenance, and limitations more accessible as well as lightweight aids for understanding and as support for interpretation.

We also found evidence that badges are **perceived as a means to build trust** by making hidden choices and processes more transparent and by making unfamiliar aspects of a visualization more accessible. However, comprehension alone was not enough for badges to foster engagement, interpretation and trust. Readers felt that **a badge needs to offer something relevant** enough to actively consult for the task at hand, e.g., by disclosing transparency when readers cared about it (Background Reading Available), by highlighting analysis processes when those were relevant to them (Aggregated Data), or by highlighting pitfalls that would otherwise go unnoticed (Missing Data). When the information conveyed through badges was not understandable, e.g., because the terminology was too complex, or the statements were too general or too obvious, badges were experienced as cluttering the display and not supporting above mentioned positive effects.

Not unexpectedly, we found **mixed evidence for aspects related to learning**. For example, participants expressed diverging views on whether badges were able to foster CRITICALITY. In S2, one participant said that badges made them more critical by highlighting limitations, while another described the opposite tension: badges increased trust because they provided more transparency, context, and information about limitations, but for that very reason also reduced the perceived need to scrutinize the visualization more critically.

Eventually, while the general audience (S1a, S1b) strongly supported CONVENTIONALITY, this support became more nuanced among professionals (S2) and especially designers (S3). Readers in S2 generally expressed that **badges could become a recognizable transparency convention**. Like fact-checking labels [38] or labels for code integrity [65] they would signal responsible and transparent practice and quality, particularly in complex visualization displays with multiple visualizations and lots of information. Designers, for their part, cautioned against becoming too prescriptive on a global scale. Instead, they saw potential for badge conventions within a specific context, such as an organization or product, where reusable and meaningful badges could be more easily established and revisited over time, allowing users to learn these conventions. *"Once such conventions were established, the absence of a badge [would] no longer be neutral"*[P2]. Absence would imply that certain information has not been disclosed, an effect observed for fact-checking labels [17].

6.2 Badges Between Transparency, Trust, and Criticality

From our findings, we assess that badges are **critical means for communicating contextual information** around visualization and that exposure to badges can for some people lead to critical thinking. These findings are different from prior studies by Burns et al. [14, 15] on textual metadata alongside visualizations, i.e. footnotes. Those prior

studies found that external information had little effect on how people read and perceive visualizations beyond this information increasing perceived thoroughness. In contrast, our findings suggest that external information was considered useful. Depending on the audience, this information included where the data came from (Data Sources Disclosed), what had been transformed or omitted (Data Filtered), which inferences require caution (Correlation ≠ Causation), or why an unfamiliar encoding should be read in a particular way (Dual-Axis Encoding). Moreover, this information was perceived to fostering trust in the visualization and data.

However, badges reveal a tension between trust and criticality, functioning as **a double-edged sword**. Rather than simply increasing trust, badges **help calibrate trust and criticality**. On the one side, badges supporting trust resonates with prior work that frames trust in visualization as a multi-dimensional construct that includes credibility, confidence, clarity, and reliability into both data and the visual representation [22, 49]. By disclosing provenance, methodological choices, and limitations, badges make explicit what readers should trust and make them aware of the concepts they should consider in their assessment. On the other side, especially warning badges can reduce confidence in a dataset or analysis while increasing trust in the author.

Badges can help explain and justify why a particular framing, time frame, or metric was chosen, but they should not reduce multiple legitimate interpretations to a single authoritative conclusion i.e., steering readers interpretation. Hence, badges are least contentious when they clarify the conditions for interpretation, and more problematic when they begin to prescribe specific conclusions.

6.3 Visualization badges as a conventional practice?

Across studies, we noticed many nuances of how badges and information were presented, strongly impacting how they were perceived and used. Consequently, we discuss implications for applying badges to visualization projects and the potential for a wider conventionalization.

A first implication is the need to **balance recognizability and design flexibility**. Readers profit from a recognizable visual language of badges, including recurring shapes, icons, color roles, and interaction conventions. Designers, however, need badges to adapt to local branding, typography, layout, and aesthetic constraints (S3). Designers suggested fixing shapes and icons throughout for recognizability while giving them flexibility over colors and fonts. So, there could be a shared visual grammar or style guides within e.g., a domain or organization.

A second implication is **how to prioritize individual badges** while not using badges as an excuse for "lazy design". Because too many badges increase visual clutter, and because only a limited amount of information can be shown before overwhelming readers [42]. Conventions should therefore include clear prioritization rules: *Which badges are most important? Which ones belong at the page or chart level? Which ones are optional?*

Likewise, we found some evidence for those **badges considered most important**. For example, some badges should be *compulsory badges*, because they speak directly to credibility, provenance, or quality e.g. (Background Reading Available). Others can be understood as optional or conditional badges, e.g., whose usefulness depends on the audience, the context e.g. (Major Finding), or the reader's prior familiarity with the visualization (Qualitative Categories). Still others may serve as ephemeral badges: temporary aids that support first encounters with a visualization, but can recede once concepts, conventions, or interaction patterns have become familiar (Interaction Linked Across Multiple Views).

A final implication is about the **credibility of who assigns a badge**. Without clear rules for applying badges and supervising their application, they risk being exploited intentionally or unintentionally, becoming void and misleading badges. This is particularly important for badges that imply quality, responsibility, or methodological rigor. In the context of promoting accountability and preventing bias in AI development, researchers created highly structured frameworks for dataset documentation, such as nutrition labels for datasets [34], datasheets [28], data statements [9], and data cards [54]. Through that lens, conventionalization is a governance problem: communities need common definitions, examples, and perhaps eventually even review or verification processes for the badges. A final concern is that **conventionalization can make the absence of badges legible**: if some projects adopt them consistently, others may have to explain *"why are you not part of the badge club?"*[P2]. While our findings capture participants' preference of badges becoming a wider used convention, lots of open questions remain about the social, legal and political processes through which such conventions and eventually standards can emerge by themselves or be actively formed.

6.4 Implications and Future Work

Visualization badges are highly related to established practices for communicating context around visualizations, such as captions [41], annotations [24], labels [13], and footnotes [30]. Badges can complement, partially replace, or become part of these techniques. For example, a warning badge in mini-mode placed next to an outlier in a visualization could expand, on click, into a detailed annotation: this would add categories and higher-level semantics to otherwise "generic" textual annotations. Likewise, in interactive environments with limited screen space, footnotes could be replaced entirely by interactive visualization badges (S1); badges can be seen as abstractions of large numbers of footnotes and disclosures. Our findings about CURIOSITY and CONCEPTS suggest that badges may successfully support visualization onboarding [58] and opportunistic learning [48]. Certain badges may eventually function as easy-to-use interfaces through which readers intuitively know what information and interactions to expect, provided that badges convey similar information and use a consistent graphical language across projects.

However, our findings also highlight cases where badges can become harmful in that they clutter the display, distract from the visualization, and induce 'badge fatigue' [21]. For example, when they state the obvious, use technical jargon, or simply when there are too many badges and too much repetition. Without specific frameworks, badges may also be misused to create an appearance of rigor and accountability without credible processes behind them, thereby miscalibrating trust.

Our findings about the potential benefits of badges therefore require further confirmation. We suggest future research building on our studies and verify, refine, and extend these findings. Future work should observe badge use in natural contexts and assess their effects on people over time. Different badges may also entail different trade-offs: for instance, financial news outlets may require different forms of disclosure than public health dashboards. Future research should therefore examine the adoption of specific badges across contexts, as well as how stakeholders—including analysts, designers, domain scientists, decision makers, and audiences—negotiate information disclosure within and beyond organizations.

6.5 Limitations

Our work is largely qualitative, examining subjective accounts of visualization designers and readers to understand their reactions to visualization badges, and the tradeoffs they perceive. We conducted three studies involving different populations, tasks and stimuli, and used qualitative coding and a visual summary scheme to synthesize findings. Because S1a and S1b used different tasks, comparing the respective results should be done with caution. S2 involved experts using real visualization environments, but ecological validity was limited because tasks were suggested, rather than defined by participants. Many of the qualities we tested assessed *perceived* benefits of badges (e.g., whether participants think they would foster trust). Direct measurement of these impacts over time is required.

7 CONCLUSION

This study found converging evidence for benefits of visualization badges. Readers notice them and interact with them, and both readers and designers consider that they can support interpretation, trust, and critical engagement, without necessarily cluttering the display. However, badges need to be carefully designed and thoughtfully integrated into a project to fully leverage their potential. We hope our studies and the supplementary material at `https://vis-badges.github.io` inspire further research and promote visualization badge adoption.

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APPENDIX OF

Mind the Badge: How Visualization Badges Impact Reading Visualizations

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A Study I

A.1 Participants

For S1a, we recruited 24 participants. 3 participants volunteered through university mailing lists, and 21 participants were recruited via Prolific. Participants who successfully completed S1a were compensated at an average rate of 15.61£/hour, with a median completion time of 19 minutes. For S1b, we invited all S1a participants back, which was followed by 18 participants (9 from the footnote-first group and 9 from the badge-first group). S1b participants were compensated at an average rate of 9£/hour, with a median completion time of 15 minutes. We recruited English speakers with at least an undergraduate degree to obtain a relatively homogeneous group in terms of educational experience. In S1a, participants self-reported gender as 58.3% men ($n = 14$) and 41.7% women ($n = 10$). Participants were between 25 and 65+ years old, and most held a bachelor's (54%) or master's degree (37.5%). Participants reported varying familiarity with visualization, with 62.5% reading charts at least a few times per month and 58.3% creating charts at least a few times per month. No participants reported color vision deficiency. For a full S1a demographics overview see Section A.2. Because S1b participants were drawn from S1a, demographics were similar but the sample was smaller, because not all participants who participated in S1a ($n = 24$) returned to S1b ($n = 18$).

A.2 Participant demographics

Tab. A1 provides an overview of the demographics of Study I.

Characteristic	Count	Percentage
Gender		
Man	14	58%
Woman	10	42%
Age		
25–34 years	6	25%
35–44 years	8	33%
45–54 years	5	21%
55–64 years	3	13%
65+ years	2	8%
Education		
Bachelor's degree	13	54%
Master's degree	9	38%
Doctorate (e.g., PhD, EdD)	2	8%
Chart reading frequency		
Rarely (few times per year)	9	38%
Occasionally (few times per month)	7	29%
Regularly (few times per week)	5	21%
Frequently (daily or almost daily)	3	13%
Chart creation frequency		
Never	3	13%
Rarely (few times per year)	7	29%
Occasionally (few times per month)	11	46%
Regularly (few times per week)	3	12%
Color vision deficiency		
No	24	100.0%

Table A1: Study I participant demographics for S1a ($n = 24$). S1b participants were a returning subset of this sample.

B Study II

B.1 Projects

B.1.1 The UK Co-benefits Atlas (Atlas)

The UK Co-benefits Atlas (Atlas) is a multipage web-based visualization atlas about the monetary co-benefits expected from implementing the Climate Change Committee's recommendations (Fig. 6, [53]). Each page in the atlas represents one of 11 co-benefits or one our of over 300 communities in the UK and features around 20 visualizations and explanations. As shown in Fig. 6, an atlas page presents a dense display of information, similar to dashboards and potentially overwhelming to novices. The atlas is intended for policymakers, businesses and other authorities to inform their decision making with respect to the NetZero transition.

B.1.2 PeaceRep (PeaceRep)

PeaceRep (PeaceRep) is a research consortium led by the University of Edinburgh and researchers peace processes to support peace building. Part of the project is engaged in creating analytical and communicative visualizations to help researchers and decision makers understand existing peace processes and inform future peace building activities based on data [8, 62]. Most projects visualize data that is to some extend uncertain and because of the sensitivity of the topic—peace and conflict—requires bespoke approaches to communication, transparency, and disclosing information.

B.2 Stage 1: Badge-selection

As a prerequisite for this study, we required a set of badges to add to the projects. To this end, we held two badge-selection workshops, one with the Atlas team and one with the PeaceRep team. Each workshop lasted approximately two hours and involved four participants from a variety of social, political and data science backgrounds (see Sec. B.2). Importantly, these participants were the people who were involved in the creation to the respective visualization project (the authors). We introduced the concept of badges and our badge catalogue [21], then used shared Miro boards containing screenshots of project pages and badges. During the first hour, each participant individually assigned badges to the projects in the miro board, which then were discussed during the second hour and eventually resulted in a set of badges for us to integrate into the projects. The following table lists the participants involved in the badge selection process.

ID	Institution	Role	Expertise	Years
E1	ECCI	Researcher, outreach	Policy communication	14
E2	ECCI	Data analyst	Data modelling	5
E3	ECCI	Lead researcher	Socio-economic analysis	4
E4	PeaceRep	Vis designer	Digital humanities	8
E6	PeaceRep	Domain researcher	Peace agreement coding	11
E7	PeaceRep	Data engineer	Peace and conflict databases	10
E8*	Both	Vis Designer	Data visualization, HCI	4

Table B1: Demographics of the workshop participants involved in the badge selection process. Participants E1–E3 and E8 participated in the first workshop. Participants E4–E8 participated in the second workshop. Note that E8participated in both workshops.

B.3 Stage 2: Badge Assignment

The second prerequisite for this study, was to affix badges to the live visualization implementations. To do so, we developed a lightweight JavaScript badge library that operationalizes the badge design space introduced in our prior work [21]. The library supports different badge placements, visual modes, and interaction affordances, which we used to implement a **badge-priority**: salient round badges for high-priority information, standard badges for regular use, and mini badges that expand on hover to reveal a tooltip and click to access more detailed information, open external links, or trigger actions such as downloading resources. The selected badges were not simply transferred from the workshop to the pages without adjustment, but were modified until they matched the projects' terminology, visual style, and communicative intent Fig. 6. In the UK Co-benefits Atlas, we integrated badges into all 11 Co-Benefit pages and all 382 Local Authority Pages. For PeaceRep, we integrated badges into three projects: *Actor Network*, *Messy Timelines*, and *Peace-Scroll*.

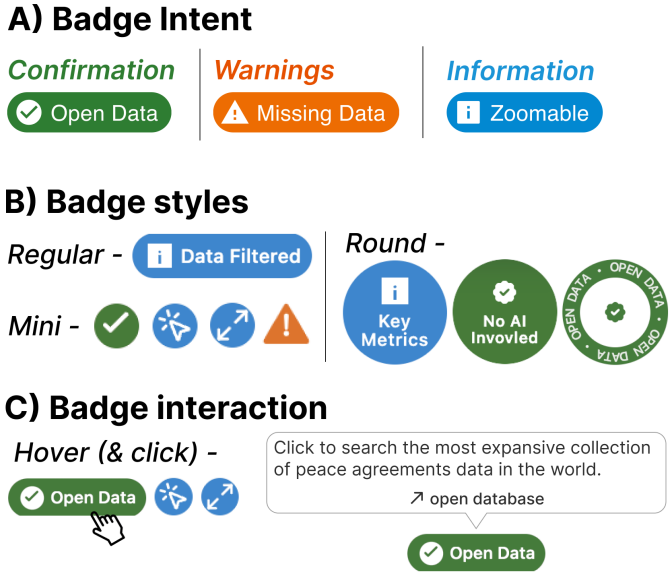


Figure B1: Visualization badges design options: A) colors usage in assigning intent from confirmation, warning, and information; B) visual styling from regular badges, mini badges and round badges; C) available interactions like hover to see the tooltips and click to open external sites.

B.4 Expert interview participants

The following table summarizes the professional backgrounds of the 12 S2 interview participants in generalized form.

ID	Role	Organization	Expertise	Years
P1	Analyst	Academia	Climate action and policy	20+
P2	Analyst	Academia	Climate action and policy	4.5
P3	Manager	Community org	Climate policy communication	9
P4	Manager	Local government	Climate action and policy	24
P5	Manager	Community org	Climate action and policy	14
P6	Researcher	Academia	Climate policy research	9
P7	Journalist	Academia	Environmental journalism	14
P8	Consultant	Private Sector	Climate governance	7+
P9	Communicator	Academia	Research communication	9
P10	Researcher	Academia	Political science	14
P11	Manager	Healthcare	Public health	20+
P12	Researcher	Academia	Geoinformatics	19

Table B2: S2 stakeholder-interview participants ($n = 12$).

B.5 Interview pages

Example pages used in the Study 2 interviews are listed below.

The Atlas examples show the Air Quality Improvements and Noise Reduction co-benefits as well as the City of Edinburgh and Scotland local authority pages:

- `ukcobenefitsatlas.net/cobenefit?cobenefit=Noise`
- `ukcobenefitsatlas.net/cobenefit?cobenefit=Air%20quality`
- `ukcobenefitsatlas.net/location?location=S12000036`
- `ukcobenefitsatlas.net/nation?nation=Scotland`

For PeaceRep, we used three projects: *Actor Network*, illustrated here through Russia's involvement in peace and transition processes; *Messy Timelines* to explore peace-process patterns; and *Peace-Scroll*, a scroll-based project, here depicting China as a third party in peace agreements.

- `vedelsbrunner.github.io/actor-network/country/RUS`
- `vedelsbrunner.github.io/v7_messy_timeline/`
- `vedelsbrunner.github.io/peace_scroll/?subset=china#`

B.6 Codes and inter-rater reliability analysis

The purpose of the qualitative coding process was to use participant responses to answer a set of predefined questions about badges, particularly concerning the level of support for each badge quality (e.g., saliency, curiosity). For each participant and each dimension (i.e., question), two coders independently applied a set of predefined codes to the responses (see Table B3). The coders then reviewed any disagreements dimension by dimension, revisited the transcripts, and discussed ambiguous cases until they reached a shared interpretation.

Dimension	Explanation	Codes
Project Expertise	Participant's prior experience with the project (Atlas or PeaceRep) before the interview.	YES | PARTIAL | NO | N/A
Saliency	Did the participant notice badges before the interviewer explicitly introduced them?	YES | UNCLEAR | NO | N/A
Curiosity	Do badges spark participants' curiosity or interest to explore the project?	YES | UNCLEAR | NO | N/A
Clutter	Concerns that badges clutter the display, increase cognitive load, or overwhelm.	YES | PARTIAL | NO | N/A
# Badges per chart	Reported acceptable number of badges per chart before overload.	INT | RANGE | UNCLEAR | N/A
Engagement	Participant reads, hovers over, or clicks badges during exploration.	YES | NO | N/A
Comprehension	Understanding of badge visual language (icon, color, label, tooltip).	YES | PARTIAL | NO | N/A
Interpretation	Influence of badges on interpretation of the visualization.	YES | UNSURE | PARTIAL | NO | N/A
Trust	Effect of badges on trust in the visualization/project.	INCREASE | PARTIAL | DECREASE | UNCLEAR | N/A
Criticality	Whether badges promote critical reflection on data/visualization.	YES | PARTIAL | NO | UNSURE | N/A
Concepts	Whether badges help learn new concepts.	YES | UNSURE | NO
Conventionality	Desire for badges as a standard in other projects.	YES | CONDITIONAL | NO | UNSURE | N/A
Badge Stance	Overall opinion of badges and their implementation.	YES | PARTIAL | NO | UNSURE | N/A

Table B3: Codes for S2.

We conducted an inter-rater reliability analysis [29] on all badge quality questions. Because the codes were intended to classify the level of support for each badge quality as negative evidence, positive evidence, mixed evidence, or no data (see subsection 2.3), we first collapsed equivalent codes: *yes/increase* as *POSITIVE* (✓), *no/decrease* as *NEGATIVE* (×), *partial/conditional* as *MIXED* (∼), and *unclear/unsure/n/a* as *UNSURE*. For the inter-rater reliability analysis, we further combined *MIXED* and *UNSURE* into one intermediate category (∼), resulting in three ordered categories—*POSITIVE*, *MIXED*, and *NEGATIVE*. While the inter-rater reliability analysis collapses *MIXED* and *UNSURE* into the same intermediate category, we distinguish these cases again in the findings and do not treat unclear responses as mixed; in fact unclear responses were excluded in the overall reported coding scheme (Sec. 2.3).

For each coding dimension (badge quality), we computed two inter-rater reliability metrics: *percent agreement* and *weighted Cohen's kappa* (κ) [29]. Percent agreement reflects the proportion of participants for whom the two coders gave the same code, while κ captures coding consistency after correcting for chance agreement (i.e., $\kappa = 1$ reflects perfect agreement, $\kappa = 0$ chance agreement, $\kappa = -1$ perfect disagreement). The weighted kappa scores were computed with the assumption that ✓ vs. × yields a disagreement of 1, while ∼ vs. either ✓ or × yields a disagreement of 0.5.

Tab. B4 reports, for each participant (P1, P2, etc.) and coding dimension (Saliency, Curiosity, etc.), the codes independently assigned by Coder 1 (C1) and Coder 2 (C2), along with the merged code (M). Percent agreements and weighted Cohen's kappas (κ) are reported in the rightmost column. Note that κ values tend to be low for dimensions where responses are highly consistent across participants, an issue known as the kappa paradox [19]. Therefore, percent agreement, together with the individual codes, should be considered alongside κ values to accurately assess coding reliability. Considering all measures, only a few dimensions showed substantial disagreement. For Engagement, low disagreement arose mainly from opposite codes on three participants, due to a misunderstanding that was fully resolved through discussion. Criticality, however, yielded numerous discrepant codes, indicating interpretive difficulty, and leading us to draw conclusions regarding this dimension with caution.

Dimension	P1			P2			P3			P4			P5			P6			P7			P8			P9			P10			P11			P12			Agr.	κ
	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M		
Saliency	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	×	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	∼	∼	✓	✓	✓	100%	1.00
Curiosity	✓	✓	✓	✓	∼	✓	∼	∼	∼	✓	✓	✓	✓	∼	∼	∼	∼	∼	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	83%	0.75
Clutter	∼	×	×	∼	∼	∼	∼	∼	∼	∼	∼	∼	∼	×	×	✓	✓	✓	×	×	×	✓	∼	∼	✓	×	×	∼	∼	∼	✓	✓	✓	∼	∼	∼	67%	0.55
Engagement	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	×	✓	×	×	✓	×	×	✓	×	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	✓	∼	✓	✓	✓	67%	0.25
Comprehension	✓	✓	✓	✓	✓	✓	∼	✓	∼	✓	∼	✓	✓	✓	✓	∼	×	∼	✓	∼	✓	∼	✓	∼	✓	∼	✓	✓	✓	✓	∼	∼	∼	✓	✓	✓	50%	0.44
Interpretation	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	✓	✓	∼	∼	∼	×	×	∼	∼	∼	∼	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	∼	✓	✓	✓	58%	0.51
Trust	✓	✓	✓	∼	✓	∼	✓	✓	✓	✓	∼	∼	∼	∼	∼	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	∼	∼	✓	✓	✓	✓	✓	✓	83%	0.75
Criticality	✓	∼	∼	✓	✓	✓	×	✓	×	✓	∼	✓	✓	✓	✓	∼	✓	∼	∼	×	∼	∼	×	×	✓	✓	✓	✓	∼	∼	✓	∼	∼	✓	✓	✓	33%	0.28
Concepts	✓	✓	✓	✓	∼	✓	✓	∼	✓	∼	∼	∼	∼	∼	∼	×	×	×	✓	✓	✓	∼	✓	∼	✓	✓	✓	✓	∼	✓	∼	×	∼	✓	∼	✓	50%	0.49
Conventionality	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	✓	∼	✓	✓	✓	∼	∼	∼	✓	✓	✓	∼	∼	∼	✓	∼	∼	✓	✓	✓	∼	✓	∼	✓	✓	✓	75%	0.66
Badge Stance	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	∼	∼	✓	✓	✓	∼	∼	∼	✓	✓	✓	✓	✓	✓	∼	✓	✓	✓	✓	✓	92%	0.86

Table B4: S2 coding; dimensions are shown in rows, participant codes are grouped by coder 1 (C1), coder 2 (C2), and the reconciled merged code used in the analysis (M), the rightmost columns report percent agreement and weighted kappa by dimension.

C Study III

C.1 Participants

For a demographics overview for this study, we point to Tab. C1.

ID	Role	Years
P1	Data journalist	5
P2	Independent designer	15+
P3	Independent designer	20+
P4	Data vis practitioner	8
P5	Independent designer	10

Table C1: S3 designer participants ($n = 5$).

C.2 Codes and Inter-rater reliability analysis

The inter-reater reliability analysis followed the same blueprint as in Sec. B.6. Codes are reported in Tab. C2, while Tab. C3 reports the independent codes assigned by Coder 1 (C1) and Coder (C3), the reconciled merged codes (M), as well as percent agreement scores and weighted kappas. As small sample sizes tend to worsen the kappa paradox [19], the phenomenon is even stronger here than in the previous study (N=5 here vs. N=12 for S2). In particular, Concepts yield a very low $\kappa = 0.17$ despite a relatively high agreement on the codes. Again, κ values need to be complemented with an examination of the individual codes and the percent agreements, which do not seem to reveal a strong disagreement for any of the dimensions coded.

Dimension	P1			P2			P3			P4			P5			Agr.	κ
	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M	C1	C2	M		
Comprehension	✓	✓	✓	✓	✓	✓	∼	✓	∼	∼	×	∼	∼	∼	∼	60%	0.61
Interpretation	∼	∼	∼	✓	∼	✓	∼	∼	∼	∼	∼	∼	∼	∼	∼	80%	0.38
Trust	✓	✓	✓	✓	✓	✓	✓	✓	✓	∼	∼	∼	∼	∼	∼	100%	1.00
Concepts	✓	✓	✓	✓	∼	✓	✓	✓	✓	✓	∼	✓	✓	✓	✓	60%	0.17
Conventionality	∼	∼	∼	∼	∼	∼	∼	✓	✓	∼	∼	∼	∼	∼	∼	80%	0.38
Badge Stance	∼	∼	∼	✓	✓	✓	∼	∼	∼	×	×	×	✓	✓	✓	100%	1.00

Table C3: S3 coding; dimensions are shown in rows, participant codes are grouped by coder 1 (C1), coder 2 (C2), and the reconciled merged code used in the analysis (M), the rightmost columns report percent agreement and weighted kappa by dimension.

Dimension	Explanation	Codes
Comprehension	Does the designer think intended users would understand the badge's visual language (icon, color, short label, tooltip) and the information it communicates about the visualization?	YES | PARTIAL | NO | N/A
Interpretation	Does the designer think badges would influence how users interpret the visualization, for example by guiding attention, supporting correct reading, adding needed context, or helping prevent misinterpretation?	YES | PARTIAL | NO |UNSURE | N/A
Build Trust	Does the designer think badges would influence users' trust in the visualization, underlying data, analysis, or project, for example by making provenance, limitations, or design choices more transparent?	INCREASE | PARTIAL | DECREASE |UNCLEAR | N/A
		YES | UNSURE | NO | N/A
Conventionality	Does the designer think badges should be used more widely across visualization projects, or see them as a desirable or standard practice beyond a single project?	YES | CONDITIONAL |UNSURE | NO | N/A
Badge Stance	Designer reports on liking the badges their concept and how they are currently implemented	YES | PARTIAL |UNSURE | NO | N/A

Table C2: Codes for S3.